

Isabelle Tingzon

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EDUCATION

KTH Royal Institute of Technology

Ph.D., Geodesy and Geoinformatics

- Research focus: Remote Sensing and Computer Vision
- Supervisors: Prof. Yifang Ban and Dr. Aleksis Pirinen

Stockholm, SWE

Feb 2026 - Present

Technical University of Munich

Beyond Fellow, AI4EO Future Lab

- Research focus: Geospatial ML for Earth Observation
- Supervisor: Prof. Xiao Xiang Zhu

Munich, DEU

Aug 2022 - Jan 2023

University of the Philippines Diliman

M.S. Computer Science

- Philippine-California Advanced Research Institutes Scholarship

Metro Manila, PHL

Aug 2015 - Jan 2018

University of the Philippines Diliman

B.S. Computer Science, Magna cum laude

- Phi Kappa Phi Honor Society

Metro Manila, PHL

Jun 2011 - Jun 2015

WORK EXPERIENCE

RISE Research Institutes of Sweden

Doctoral Researcher in Machine Learning

- Conduct research on robust and data-efficient ML for multimodal Earth Observation data under the project ML-Earth, funded by the Swedish Foundation for Strategic Research (SSF), as part of the Deep Learning Research Group at RISE.

Lund, SWE

Feb 2026 - Present

The World Bank

Geospatial Data Scientist, Junior Professional Associate

Disaster and Climate Risk Data Fellow

- Developed data-efficient workflows for building and parcel segmentation from high-resolution Maxar and Airbus satellite imagery using SAM3, DINOv3, UMAP, and HDBScan as part of the Spatial Parcel Atlas for Resilient Cities (SPARC) initiative to support investment, land administration, and service delivery decisions in African cities.
- Developed the [Disaster-FCV Scan](#) , an open-source, globally applicable Python library for the rapid mapping of multi-hazard and conflict exposure at subnational scales to identify assets, i.e. populations, infrastructure, cropland, facing compounded risks.
- Led AI-driven climate resilience initiatives in the Caribbean, as featured in the UNFCCC TEC Information Note on “AI for Climate Action in Developing Countries: Opportunities, Challenges, and Risks”, launched at COP29.
- Leveraged computer vision models (CNNs, ViTs) to generate nationwide building stock classification maps in Caribbean small island developing states (SIDS), made accessible via the [World Bank Data Catalog](#) , using drone, LiDAR, satellite, and street view imagery in support of government programs for climate-resilient housing and infrastructure.
- Delivered technical training on AI-driven workflows to government staff, community mappers, and disaster responders as part of local capacity-building initiatives for climate resilience in the Caribbean.
- Co-authored a successful grant proposal that secured 50K USD to scale our AI for climate resilience pilot in the Caribbean.

Washington DC, USA

Jan 2025 - Jan 2026

Feb 2023 - Dec 2024

United Nations Children’s Fund (UNICEF)

Data Science Consultant

- Led our AI-enabled global school mapping initiative, leveraging computer vision and Earth Observation data in support of Giga, a UNICEF-ITU initiative aimed at achieving universal school connectivity.
- Built an end-to-end pipeline for large-scale school mapping using high-resolution Maxar satellite imagery, weakly supervised deep learning models (ViTs, CNNs), and explainable AI (xAI), mitigating costly bounding box annotation.
- Deployed models in 10+ countries across Africa and Central Asia, generating national-scale maps of school location predictions using Dell’s high-performance computing (HPC) infrastructure.
- Co-developed a human-in-the-loop model validation dashboard prototype using Python Dash to assist government partners in discovering non-geolocated school infrastructure and improving the quality and completeness of official school location data.
- Facilitated knowledge transfer by presenting technical concepts to UNICEF country offices, government agencies, and ministries of education through comprehensive reports, training sessions, and slide deck presentations.
- Presented at the AAAI 2025 Special Track on AI for Social Impact (oral, top 5% of 469 submissions)
- Awarded Best Innovative Technology Application Award at the IEEE Connecting the Unconnected (CTU) Summit 2024.

Barcelona, ESP

Oct 2023 - Jan 2025

Thinking Machines Data Science

GeoAI Team Lead, Machine Learning Researcher

Metro Manila, PHL

Mar 2018 - Jul 2022

- Led a team of 7 ML researchers and engineers in building our GeoAI product, leveraging ML and computer vision for building delineation and classification, land use and land cover classification, and socioeconomic class classification from high-resolution satellite images to support high-CAPEX decision-making by major telcos across Southeast Asia.
- Generated high-resolution, nationwide poverty maps in the Philippines using ML/DL, satellite imagery, social media data, and volunteered geographic information to support targeted humanitarian interventions by UNDP Philippines.
- Accelerated the detection of 350+ informal migrant settlements in Colombia during of the Venezuelan mass migration crisis using ML and time-series Sentinel-2 satellite images; model predictions were ground-validated by local contributors in Colombia using mobile apps and shared with local NGOs and UN agencies, enabling more efficient, targeted humanitarian operations. Collaborators: iMMAP Colombia and Premise Data.
- Conducted field visits in Bogota and La Guajira to assess the living conditions of Venezuelan migrants situated in informal settlements in Colombia and developed grant proposals resulting in USAID-funded AI for humanitarian action projects worth 55K USD. Collaborators: UNICEF Colombia and iMMAP
- Co-authored successful grant proposals under the Sustainability Team, resulting in two AI for climate action projects worth 150K+ USD funded by the CCAI Innovation Fund and the Lacuna Fund.

TECHNICAL SKILLS

Data Science Tools: Python, Pytorch, SQL, BigQuery, Jupyter, Pandas, NumPy, SciPy, Scikit-learn, Weights & Biases

Geospatial Tools: QGIS, GDAL, GeoPandas, Rasterio, Google Earth Engine (GEE)

Cloud Computing: GCP, AWS, Azure

AWARDS

1st Place Innovative Technologies, IEEE Connecting the Unconnected (CTU) Challenge 2024	Dubai, UAE	Oct 2024
Women in AI (WAI) Asia and the Pacific (APAC) 2024: AI for Social Good Award	Sydney, AUS	Jun 2024
Best Poster Runner-up, International Future Lab AI4EO Symposium 2022	Munich, DEU	Oct 2022
Best Paper, NeurIPS 2020 Machine Learning for the Developing World (ML4D) Workshop	Virtual	Dec 2020
Best Paper and Best Presenter at KDD 2020 Fragile Earth Workshop	Virtual	Jul 2020

VOLUNTEER EXPERIENCE

Tutorials Co-lead, Climate Change AI - Core Team 2020 - 2026

- Manage the CCAI tutorial series; initiated and co-organize the Tutorials Track at the Tackling Climate Change with Machine Learning Workshop at major ML conferences, leading to the development of 25+ tutorials since 2021.

AI/ML Tech Lead, Women Who Code Manila 2017 - 2024

- Facilitated monthly study group sessions; co-organized “Filipinas in Tech 2022: Women in Data Science”, featuring leading Filipina data scientists as speakers and panelists, with a focus on career development and special topics.

SELECTED PUBLICATIONS

See my [Google Scholar](#) for the complete list.

- [Tingzon, I., Ozturk, U. C., & Dotu, I. \(2025\).](#) Large-scale school mapping using weakly supervised deep learning for universal school connectivity. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(27), 28449-28457. doi:10.1609/aaai.v39i27.35067
- [Tingzon, I., Cowan, N. M., & Chrzanowski, P. \(2024\).](#) Mapping housing stock characteristics from drone images for climate resilience in the Caribbean. *Environmental Data Science*, 3, e29. doi:10.1017/eds.2024.46
- [Tingzon, I., Cowan, N. M., & Chrzanowski, P. \(2023\).](#) Fusing VHR post-disaster aerial imagery and LiDAR data for roof classification in the Caribbean. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2023, pp. 3740-3747. doi:10.1109/ICCVW60793.2023.00402
- [Tingzon, I., Miraflor, J.M., Zhu, X.X., & Kochupillai, M. \(2023\).](#) Towards impactful applications of AI4EO in the Global South. *2023 Joint Urban Remote Sensing Event (JURSE)*, pp. 1-4. IEEE. doi:10.1109/JURSE57346.2023.10144197
- [Fatehkia, M., Tingzon, I., Orden, A., Sy, S., Sekara, V., Garcia-Herranz, M., & Weber, I. \(2020\).](#) Mapping socioeconomic indicators using social media advertising data. *EPJ Data Science*, 9(1), 22. doi:10.1140/epjds/s13688-020-00235-w
- [Tingzon, I., Orden, A., Sy, S., Sekara, V., Weber, I., Fatehkia, M., Garcia-Herranz, M. & Kim, D. \(2019\).](#) Mapping poverty in the Philippines using machine learning, satellite imagery, and crowd-sourced geospatial information. *The International Archives of Photogrammetry, Remote Sensing, and Spatial Information Sciences*, XLII-4/W19, 425–431. doi:10.5194/isprs-archives-XLII-4-W19-425-2019

Integrating Multi-Hazard and Conflict Exposure Information for Disaster Risk Management in FCV Settings

- World Conference on Humanitarian Studies 2025 Session on “Disaster Risk Reduction in Fragile, Conflict-Affected, and Vulnerable (FCV) Contexts: Strategies for Protracted Crises”. 16 Oct 2025. Istanbul, Turkey. Virtual. (contributed talk)

Bridging Data Gaps: Harnessing AI and Earth Observation for Climate Resilience and Universal Connectivity

- UNESCO Chair Webinar Series on Disaster Risk Reduction and Resilience Engineering. 25 Sep 2025. Virtual. (invited talk)
- OECD Laboratory for Geospatial Analysis Technical Workshop on Building Data Together: Building-related Open Microdata. 24 Jan 2025. Paris, France. Virtual. (invited talk)
- Applied Machine Learning Days (AMLD), Swiss Federal Institute of Technology Lausanne (EPFL). Invited Panelist and Presenter for the tracks “Accelerating Climate Change Action Through Machine Learning” and AI & Space: Algorithms for Socio-economic Benefits”. 25-26 Mar 2024. SwissTech Convention Center. Lausanne, Switzerland. (invited talk)

AI for Climate Action in Developing Countries: Opportunities, Challenges and Risks

- Infrastructure Digital Learning Event. 30 Sep 2025. World Bank Headquarters, Washington DC, USA. (invited talk)
- 6th Indonesia’s SDGs Annual Conference 2023. Co-presented with Priya Donti. 7 Nov 2023. Virtual. (invited talk)
- 9th Annual Public Policy Conference. 19 Sep 2023. Asian Institute of Management, Metro Manila, Philippines. (invited talk)

Large-scale School Mapping using AI and Satellite Images for Universal School Connectivity

- Erasmus Mundus Giga Meet. 28 Nov 2024. Universitat Pompeu Fabra, Barcelona, Spain. (invited talk)
- AI4Development Conference. 28 Oct 2024. Norrskén House, Barcelona, Spain. (invited talk)
- IEEE Connecting the Unconnected Summit 2024. 15 Oct 2024. Dubai, UAE. Virtual. (invited talk)

Mapping Housing Stock Characteristics from Drone Images for Climate Resilience in the Caribbean

- Anticipating the Crisis: Building Resilience for Communities in the Face of a Changing Climate. 21 Jan 2026. The National Autonomous University of Mexico (UNAM), Mexico City, Mexico. (invited talk)
- World Bank Land Conference 2025 Session on “Building Resilience through Effective Land Administration, Geospatial Systems and Digital Twins”. 7 May 2025. Washington DC, USA. (invited talk)
- Climate Informatics 2024. 22 Apr 2024. Alan Turing Institute, BMA House, London, UK. Virtual. (contributed talk)
- Guest Lecturer at the Columbia Climate School. 20 Mar 2024. Columbia University, New York City, USA. Virtual. (invited talk)
- Lightning Talk at the American Geographical Society Fall Symposium, Geography 2050: The Changing Map of Risk, Hazards, & Finance. 16 Nov 2023. Columbia University, New York City, USA. (contributed talk)
- Invited Panelist for the Session on “Harnessing Digital Innovation for Inclusive and Sustainable Risk Management” at the GFDRR Partnership Days 2023. 22 May 2023. The World Bank, Washington DC, USA.

Detecting Informal Migrant Settlements using Machine Learning, Satellite Images, and Mobile Apps

- Lightning Talk at the Geo for Good Summit 2020. 20-22 Oct 2020. Virtual. (contributed talk)
- KDD Fragile Earth Workshop 2020. 24 Aug 2020. Virtual. (contributed talk)

Mapping Poverty in the Philippines using Machine Learning and Satellite Imagery

- Invited Panelist at Practical Machine Learning for Developing Countries Workshop ICLR 2021. Virtual. 8 May 2021.
- Workshop on Big Data in Official Statistics for Measuring Digital Economy and Sustainable Development. 9 May 2019. National Administrative Department of Statistics, Bogota, Colombia. (invited talk)
- Python Conference Asia Pacific (PyCon APAC) 2019. 24 Feb 2019. iAcademy, Metro Manila, Philippines. (contributed talk)
- AI for Social Good Workshop. Qatar Computing Research Institute, Hamad Bin Khalifa University. 17 Feb 2019. Ar-Rayyan, Qatar. (invited talk)

ACADEMIC SERVICES

Reviewer/Meta-reviewer

- ML4RS at ICLR’26; AAAI’26; Tackling Climate Change with Machine Learning Workshop at NeurIPS’25, ICLR’23, NeurIPS’22, & NeurIPS’21; Climate Informatics 2024; Frontiers in Big Data - Data Analytics for Social Impact; IEEE Latin America Transactions Special Issue on AI and Sustainability; Philippine Journal of Science.

Workshop Organization

- AI for Climate and Conservation (AICC) Workshop at EurIPS’25; Tutorials Track at the Tackling Climate Change using Machine Learning Workshops at NeurIPS’25, ICLR’24, NeurIPS’23, ICLR’23, NeurIPS’22, & NeurIPS’21.